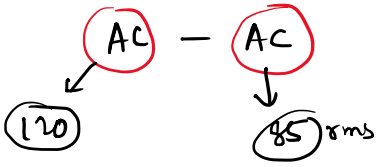
AC-AC Voltage Controllers

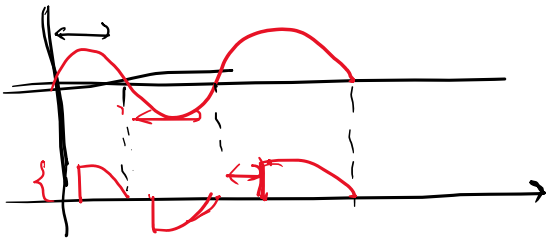
Friday, May 1, 2026 8:35 PM



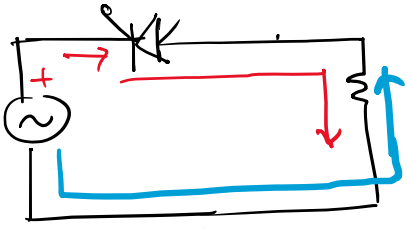
Ac - AC

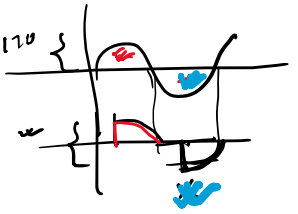
1) $V_{o,rms}$

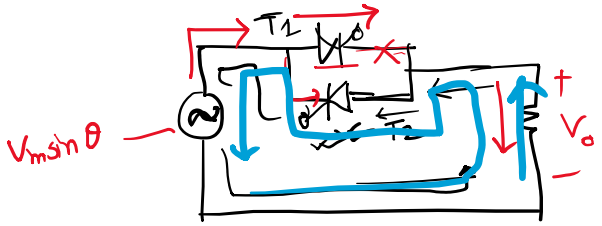
2) frequency (Cycloconverter)

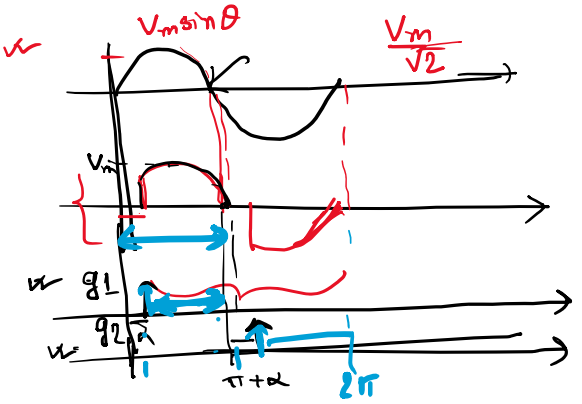


#









$$V_{o,rms}^2 = \frac{1}{\pi}\int_{\alpha}^{\pi} (V_m \sin\theta)^2 d\theta$$

$$= \frac{V_m^2}{\pi}\int_{\alpha}^{\pi} \sin^2\theta\, d\theta = \frac{V_m}{2\pi}\cdot\int_{\alpha}^{\pi}(1-\cos 2\theta)\, d\theta$$

$$= \frac{V_m^2}{2\pi}\left[\theta - \frac{\sin 2\theta}{2}\right]_{\alpha}^{\pi}$$

$$= \frac{V_m^2}{2\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)$$

$$V_{o,rms} = \frac{V_m}{\sqrt{2}}\sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

$\alpha = 0$

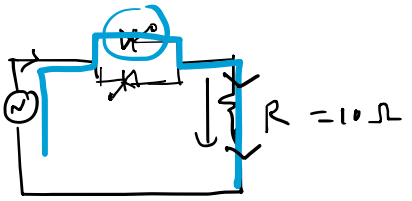
$$V_{o,rms} = \frac{V_m}{\sqrt{2}}$$

$\alpha = 0$

$$V_{o,rms} = \frac{V_m}{\sqrt{2}}$$

120V (rms), 60 Hz

$\alpha = 90 = \pi/2$



$\frac{V_m}{\sqrt{2}} = 120V$

V_o (rms)

$$V_o = \frac{V_m}{\sqrt{2}} \sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$

$$= 120 \sqrt{\frac{1}{\pi}\left(\pi - \frac{\pi}{2} + \frac{\sin \pi}{2}\right)}$$

$$= 120 \times \sqrt{\frac{1}{2}} = \frac{120}{\sqrt{2}}$$

$$= 84.85$$

Input pf

$$PF = \frac{P}{V_{rms} \cdot I_{rms}}$$

$$= \frac{719.95W}{120 \times 8.485}$$

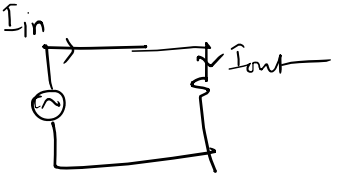
$$= \frac{1}{\sqrt{2}} = 0.707$$

$$P = I_{rms}^2 \cdot R$$

$$= 8.485^2 \times 10 = 719.95W$$

$$I_{rms} = \frac{V_{o,rms}}{R}$$

$$= 8.485\ A$$



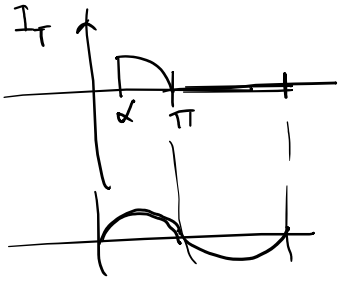
I_T (avg):

$$I_T = I_o = \frac{V}{R} \quad (\alpha - \pi)$$

$$V = V_m \sin\theta$$

$$= 120\sqrt{2} \sin\theta$$

$$I_T(\theta) = \frac{V}{R} = \frac{120\sqrt{2} \sin\theta}{R}$$



$V_m = 120\sqrt{2}$

$$I_T(avg) = \frac{1}{2\pi} \int_{\alpha}^{\pi} I_T(\theta)\, d\theta$$

$$= \frac{1}{2\pi} \cdot \int_{\alpha}^{\pi} \frac{120\sqrt{2} \sin\theta}{R}\, d\theta = \frac{120\sqrt{2}}{2\pi R} \int_{\alpha}^{\pi} \sin\theta\, d\theta$$

$$= \frac{1}{2\pi} \cdot \int_{\alpha}^{\pi} \frac{120\sqrt{2}}{R} \sin\theta \, d\theta \qquad = \frac{120\sqrt{2}}{2\pi R} \int_{\alpha}^{\pi} \sin\theta \, d\theta$$

$$= \frac{120\sqrt{2}}{2\pi R} (\cos\alpha - \cos\pi)$$

$$= \frac{120\sqrt{2}}{2\pi R} (\cos\alpha + 1)$$

$$= \frac{120\sqrt{2}}{2\pi \times 10} \times 1$$

$$= 2.7\,A$$

I_T (rms)

$$I_T(\theta) = \frac{120\sqrt{2}\sin\theta}{R} \qquad (\alpha - \pi)$$

$$I_T^2(rms) = \frac{1}{2\pi} \int_{\alpha}^{\pi} I_T^2(\theta) \, d\theta =$$

$$= \frac{1}{2\pi} \cdot \frac{(120\sqrt{2})^2}{10^2} \cdot \int_{\alpha}^{\pi} \sin^2\theta \, d\theta$$

$$I_T^2(rms) = \frac{1}{2\pi} \frac{(120\sqrt{2})^2}{10^2} \cdot \frac{1}{2} \cdot \left[\pi - \alpha + \frac{\sin 2\alpha}{2}\right]$$

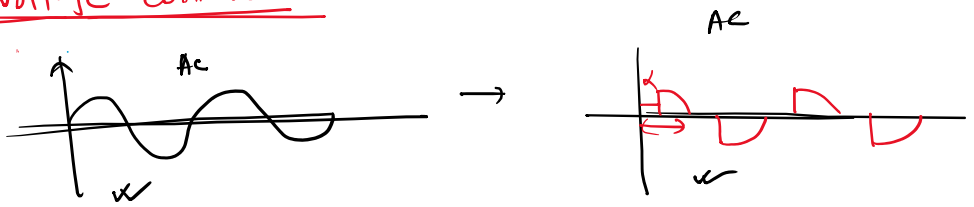
$$I_T^2(rms) = \frac{120^2}{2\pi \times 10^2} \cdot \left[\pi - \alpha + \frac{\sin 2\alpha}{2}\right]$$

$$I_T(rms) = \frac{120}{10\sqrt{2}} \cdot \sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)} \qquad \alpha = \pi/2$$

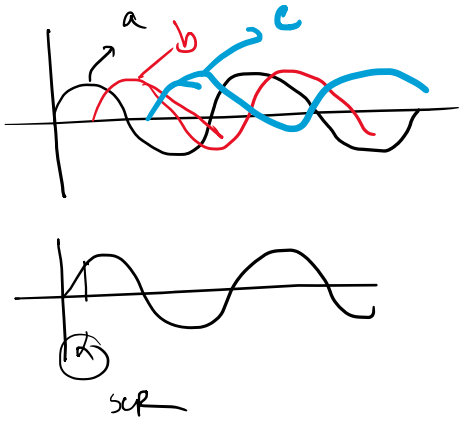
$$= \frac{120}{10\sqrt{2}} \sqrt{\frac{1}{\pi}\left(\pi - \frac{\pi}{2}\right)}$$

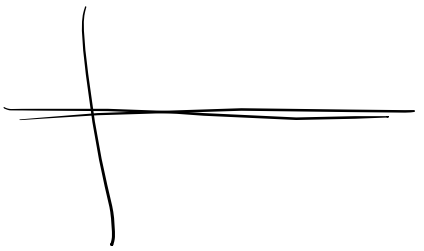
$$= \frac{120}{10 \times 2} = 6\,A$$

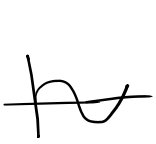
34 Ac Voltage Controller

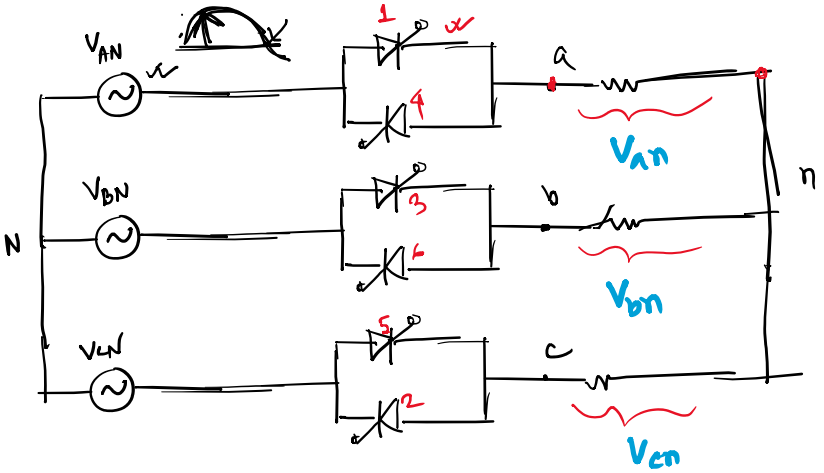


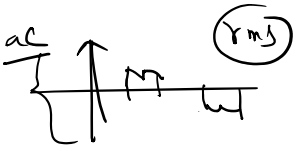
$$V_{0,rms} = \frac{V_m}{\sqrt{2}} \sqrt{\frac{1}{\pi}\left(\pi - \alpha + \frac{\sin 2\alpha}{2}\right)}$$







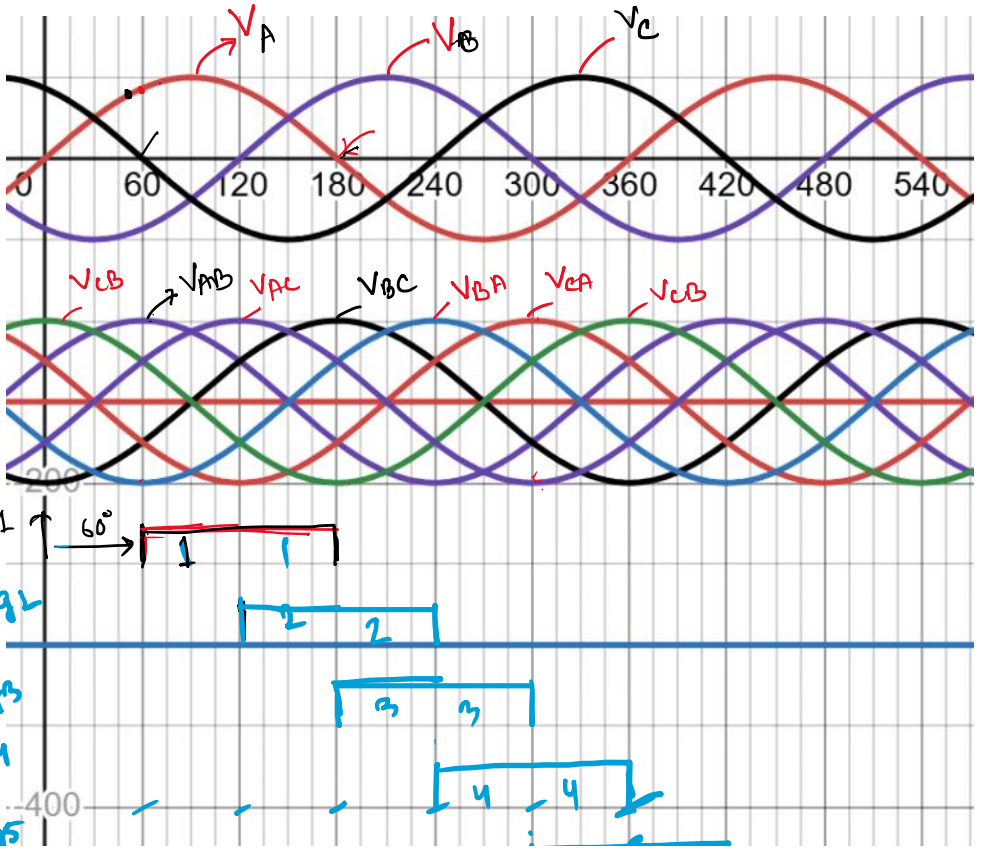




##

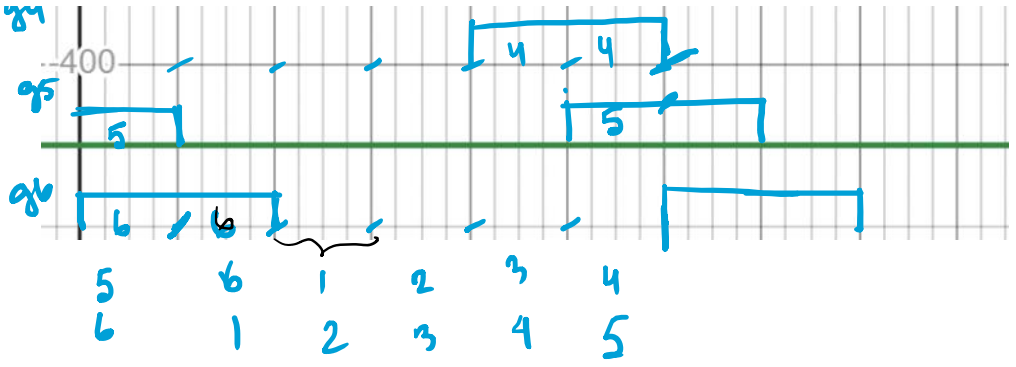
phase

line

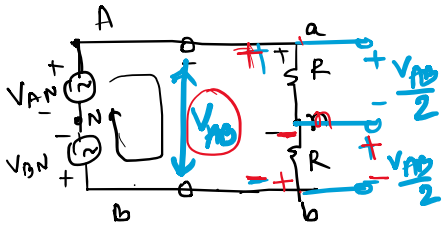


$$V_{AB} = V_A \angle 30^\circ$$

$\alpha = 60^\circ$



$\alpha = 60^\circ$ $\quad 60^\circ < \theta < 120^\circ$ $\quad$ S6, S1



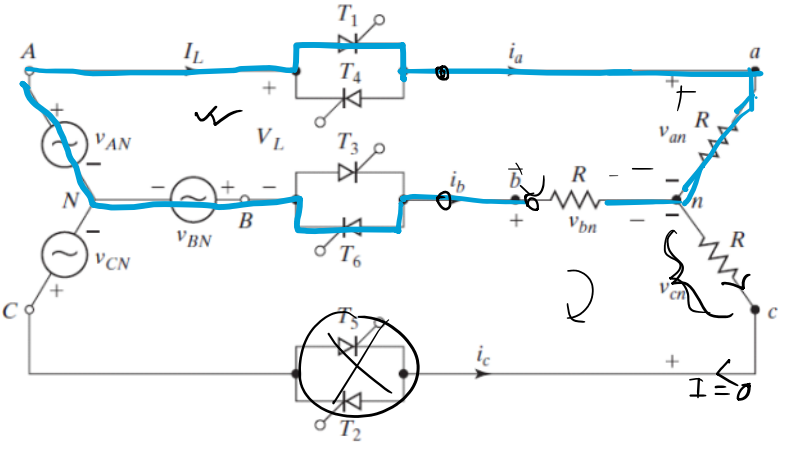
$$-V_{AN} + v + V_{BN} = 0$$

$$\Rightarrow v = V_{AN} - V_{BN} = V_{AB}$$

$$\Rightarrow V_{an} = \frac{V_{AB}}{2} \qquad V_{nb} = \frac{V_{AB}}{2}$$

$$\Rightarrow V_{bn} = -\frac{V_{AB}}{2} \qquad \Rightarrow V_{cn} = 0$$

$$V_{ab} = V_{AB} \qquad V_{bc} = -\frac{V_{AB}}{2} \qquad V_{ca} = -\frac{V_{AB}}{2}$$



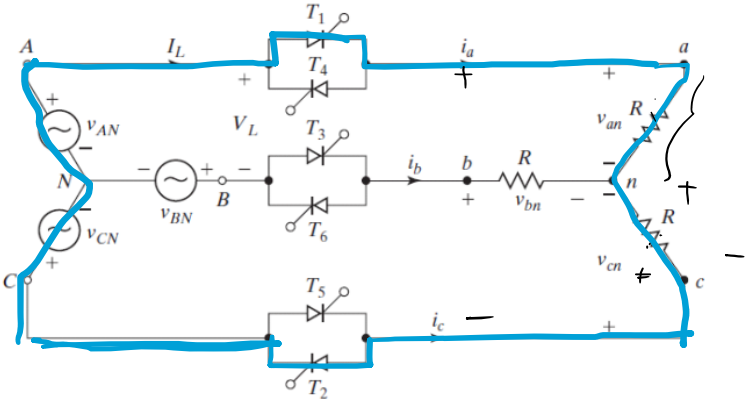
$120^\circ < \theta < 180^\circ$ $\quad$ S1, S2

$$V_{ac} = V_{AC}$$

$$V_{an} = \frac{V_{AC}}{2} \qquad V_{bn} = 0$$

$$V_{cn} = -\frac{V_{AC}}{2}$$

$$V_{ab} = \frac{V_{AC}}{2} \qquad V_{bc} = \frac{V_{AC}}{2}$$

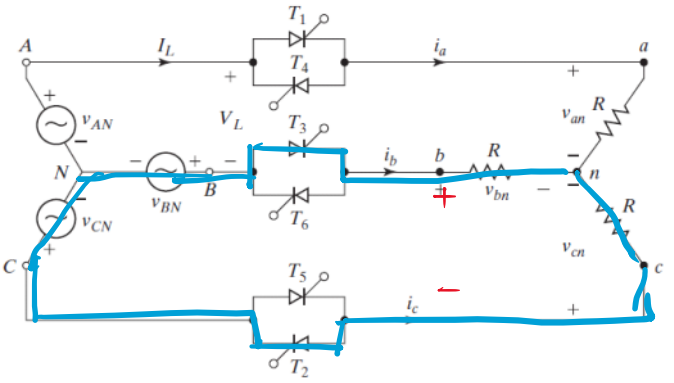


$180^\circ < \theta < 240^\circ$ $\quad$ S2, S3

$V_{bc} = V_{BC}$

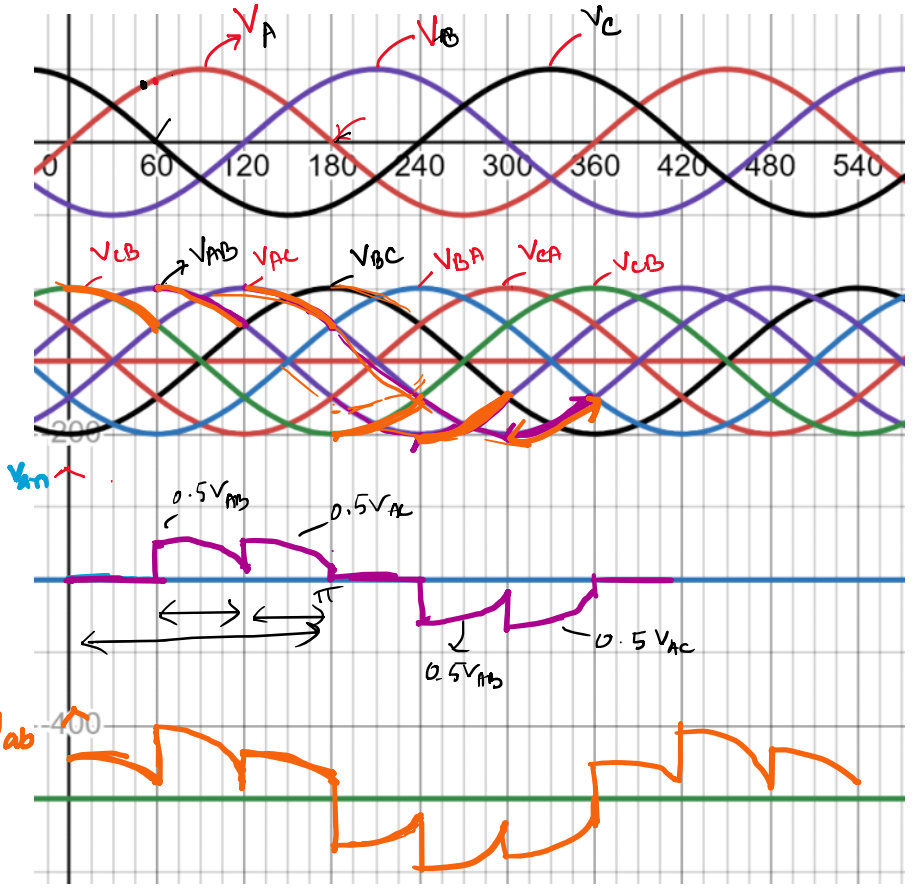
$V_{bn} = \frac{V_{BC}}{2}$ $V_{cn} = -\frac{V_{BC}}{2}$ $V_{an} = 0$

$V_{ab} = -\frac{V_{BC}}{2}$ $V_{ca} = -\frac{V_{BC}}{2}$



V_{an}

	V_{an}	V_{ab}
0–60 →	0	$-V_{BC}/2$
60–120 →	$V_{AB}/2$	V_{AB}
120–180 →	$V_{AC}/2$	$V_{AC}/2$
180–240 →	0	$-V_{BC}/2$
240–300 →	$V_{AB}/2$	V_{AB}
300–360 →	$V_{AC}/2$	$V_{AC}/2$



$-\frac{V_{BC}}{2} = \frac{V_{CB}}{2}$

$V_{an(rms)}^2 = \frac{1}{2\pi}\int_0^{2\pi} v_{an}^2(\theta)\,d\theta$

V_{AN}